Business Intelligence Final Exam Cram Pack

SECP3213 · Universiti Teknologi Malaysia

Alteryx • Tableau • Theory (Topics 1–5)

Zero-to-ready study pack · built 12 July 2026

Contents: Master Plan · Section B (Alteryx) · Section C (Tableau) · Section A (Theory) · Predicted Mock · Answer Scheme

START HERE — BI Final Exam Cram Plan (from zero)

SECP3213 Business Intelligence

You told me you're starting from zero with almost no time. Good news: **this exam is one of the most predictable you'll ever sit**, and I've reverse-engineered the whole thing. Follow this plan and you can walk in genuinely ready.

The one insight that changes everything

The datasets sitting in your exam folder — `ProductOrders_2010–2013.xlsx` + `Product_Returned.xlsx` — are the **exact same files** used in last year's real final paper, which is also in your folder. The exam is a **known quantity**:

- **Section B (Alteryx)** = build one ETL pipeline (Input → Join → Union → fix nulls → export `.hyper`).
- **Section C (Tableau)** = answer ~10 questions on that data (%, profit, forecast, CI, scatter, returns, retention).
- **Section A** = short theory (MCQ/definitions), Topics 1–5.

You are not learning "all of Alteryx and Tableau." You are learning **one workflow and ~10 chart moves.** That's it.

Exam facts (from your instruction photo)

Date/Time	**Tue 14 July 2026, 9:00 AM – 12:00 PM** (3 hours)
Venue	N28, Level 5
Weight	30% of the course
Software	**Alteryx & Tableau** installed on your laptop (Power BI **not** needed for hands-on)
Part 1	Section A — 20 min, **paper booklet, no laptop**
Part 2	Sections B & C — 2h30, **on your laptop**, save into a `MATRIC_FULLNAME` folder

> **Note:** your photo dates the exam **14 July** — that's likely **more** time than the "24 hours" you thought. Confirm your real date. The plan below front-loads the essentials so it works even in a tight 24h window.

Marks reality: Sections B + C are ~**90%** of the paper and are *pure hands-on skill*. That is where your hours go. Section A is ~10% and is easy points from the crib sheet.

Your files (in this folder)

File	What it's for	Priority
`00_START_HERE_Master_Plan.md`	This plan	Read first
`01_Section_B_Alteryx_Walkthrough.md`	Build the 50-mark Alteryx pipeline, click-by-click	● Highest
`02_Section_C_Tableau_Walkthrough.md`	All 10 Tableau questions solved	● Highest
`03_Section_A_Theory_CribSheet.md`	Topics 1–5 + past MCQ answer key	● Medium
`04_Predicted_Mock_Exam.md`	Fresh practice paper (self-test)	● After learning
`05_Answer_Scheme.md`	Grade yourself + trap notes	● With the mock

The schedule (≈9–10 focused hours + sleep)

Adjust start times to your reality. Take a 10-min break each block.

BLOCK 1 — Alteryx, learn it (2.5 h) ●

1. Read `01_Section_B` once, slowly (20 min).
2. Open Alteryx. Build the pipeline **following every step** (45 min).
3. Delete it. Build it **again** from the checklist only (30 min).
4. Build it a **third** time, blind, timed — aim < 10 min (25 min).

> Outcome: you can produce `ProductOrders...hyper` from a blank canvas without notes.

BLOCK 2 — Tableau, learn it (3 h) ●

1. Read `02_Section_C` once (20 min).
2. Load the `.hyper` you just made. Do **Q1–Q10** following the steps (90 min).
3. Redo the 5 hardest (Q1 %, Q4 CI, Q5 forecast, Q6 scatter, Q10 combined set) **blind** (45 min).

> Outcome: each question type is muscle memory. These repeat every year.

BLOCK 3 — Theory (1 h) ●

1. Read `03_Section_A` (25 min).
2. Memorize the **past-paper MCQ answer key** + the 5-V's, ETL/ELT, star/snowflake, the 4 analytics types, scorecard vs dashboard (25 min).
3. Self-quiz (10 min).

BLOCK 4 — Full mock, timed (2 h) ●

1. Section A closed-book, 15 min.
2. Sections B & C on laptop, ≤ 90 min, notes closed.
3. Grade with `05_Answer_Scheme`. Note every miss.

BLOCK 5 — Patch weak spots (1 h)

Re-drill whatever the mock exposed. Usually: the Union L/J trick, the null-Status formula, the 95% CI band, and the combined-set retention question.

SLEEP. Then EXAM MORNING (30–45 min warm-up)

- Build the Alteryx pipeline blind, once.
- Skim the crib MCQ key.
- Do 2 Tableau questions to warm your hands.

Exam-day playbook

Bring: laptop + **charger**, matric card, subject registration slip. Arrive **15 min early**. Phone/bag to the back.

When Part 2 starts: 1. **First 2 minutes:** make the folder `MATRIC_FULLNAME` on your Desktop. Download the datasets into it. Save your Alteryx workflow into it **now**. 2. **Do Section B fully** (it feeds Section C). Export the `.hyper`. **Ctrl+S often.** 3. **Load the `.hyper` in Tableau.** Answer each question on its **own worksheet renamed Q1, Q2, …** Export as `.twbx` into your folder. 4. **Never leave anything blank** — every task and worksheet is graded independently, with partial credit. A half-right answer still scores.

Order of attack in Section C: do the easy counts first (Q7, Q8, Q9 → guaranteed marks), then the medium ones (Q1, Q2, Q3, Q6), then the "fancy" ones (Q4 CI, Q5 forecast, Q10 sets).

Mindset

You don't need to be a BI expert by Tuesday. You need to reproduce **one Alteryx pipeline** and **ten Tableau moves**, and remember **~15 definitions**. That is a *completely achievable* target in the time you have. Do the reps, trust the walkthroughs, and you'll be one of the calmer people in that lab.

You've got this. Now open `01_Section_B` and start Block 1.

SECTION B — ALTERYX (≈50 marks)

Zero-to-Hero Walkthrough — SECP3213 Business Intelligence Final

> **What this is:** The exact workflow you will almost certainly build in the exam, step by step, on the real datasets already in your folder. Last year's paper used the **identical** files (`ProductOrders_2010–2013.xlsx` + `Product_Returned.xlsx`). Learn this one workflow cold and you own Section B.

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0. The 90-second mental model

Alteryx = you drag **tools** (little boxes) onto a **canvas**, connect them left-to-right with arrows, and data flows through like a pipeline. Each tool does ONE thing. You configure the selected tool in the **left panel**.

The exam workflow is a straight pipeline of 5 ideas:

```
[Input: 4 order files] ─┐
                        ├─►[Join on Order ID]─►(L + J)─►[Union]─►[Formula: fix null
Status]─►[Output .hyper]
[Input: returns file] ──┘
```

That's it. Six tools. Everything below is just *where to click*.

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1. The Alteryx screen (memorize these 4 zones)

Zone	Where	What it does
Tool Palette	Top ribbon (Favorites, Preparation, Join, etc.)	Where you grab tools
Canvas	Big middle area	Where you drop & connect tools
Configuration	Left panel	Settings for whichever tool is selected
Results	Bottom panel	Shows the data after you Run

Golden rules: - **Click a tool once** → its settings appear on the left. - **Drag from a tool's output triangle** to another tool's input triangle = connect them. - **Run** = the blue ► button top-right (or **Ctrl+R**). Always Run to see your data update. - Every tool has **anchors** (triangles): left = input, right = output.

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2. FIRST THING IN THE EXAM (do this before anything)

1. On your Desktop, **create a folder** named with your matric no. + full name (the paper demands this), e.g. `A20EC0001 MOHAMMED ALSAKKAF`.
2. **Download the exam datasets** from e-Learning into that folder.

3. Open **Alteryx Designer**. **File → Save As** → save the blank workflow into that folder as `SectionB.yxmd` **now**, and press **Ctrl+S** after every step. (Never lose work.)

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3. THE WORKFLOW — step by step

TASK 1 — Read all 4 ProductOrders files with ONE Input tool (≈8 marks)

The trick: a **wildcard** `*`. One Input Data tool can read `ProductOrders_2010`, `_2011`, `_2012`, `_2013` at once because only the year differs.

1. Tool Palette → **Favorites** (or **In/Out**) → drag **Input Data** onto the canvas.
2. Left panel → **Connect a File or Database** → browse to your folder → pick **`ProductOrders_2010.xlsx`**.
3. A sheet box pops up → choose **`Sheet1`** → OK. (You now see 2010 data in the config preview.)
4. **Now make it a wildcard.** In the file-path box on the left, find the filename and change: `...\ProductOrders_2010.xlsx|||Sheet1` to `...\ProductOrders_*.xlsx|||Sheet1` (Replace the `2010` with `*`. Keep the `|||Sheet1` part.)
5. If Alteryx asks about multiple files → accept / choose **"Read all"**.
6. *(Recommended)* Tick **"Output File Name as Field"** near the bottom of the config → this adds a column showing which year each row came from. Good insurance, not required.
7. Drag a **Browse** tool (Favorites) onto the Input's **output** anchor to preview. Press **Ctrl+R** to Run. You should see rows from all four years stacked together.

Check: Results row count should be roughly 4 years of orders (thousands of rows), not just 2010.

> **If the wildcard fights you (fallback for marks):** drag **4 separate Input Data tools** (one per file) and feed all four into a **Union** tool (Join palette) set to *Auto Config by Name*. Same result. The wildcard is the "elegant" answer worth full marks, but a working union still gets you most of them.

TASK 2 — Add a second Input for the returns file (≈5 marks)

1. Drag a **second Input Data** tool onto the canvas (below the first).
2. Browse → pick **`Product_Returned.xlsx`** → **Sheet1** → OK.
3. *(Optional)* add a **Browse** + Run. You'll see just two columns: **Order ID** and **Status** (all "Returned").

Check: This table has only `Order ID` and `Status`.

TASK 3 — Join orders with returns on Order ID (≈15 marks)

The **Join** tool matches rows from two tables using a common key. Here the key is **Order ID**.

1. Tool Palette → **Join** category → drag the **Join** tool onto the canvas.
2. Connect **ProductOrders output → Join's LEFT (top) input**. *(Orders must be the Left input — this matters for Task 4.)*
3. Connect **Product_Returned output → Join's RIGHT (bottom) input**.
4. Click the Join tool. In Configuration: - Left side dropdown: **Order ID** - Right side dropdown: **Order ID** - (Leave "Join by Specific Fields" selected.)

5. *(Optional tidy)* Scroll the field list below and **untick** the right-hand duplicate `Right_Order ID` so it won't clutter the output.
6. **Ctrl+R** to Run.

Understand the 3 outputs (this is the whole exam trick):

Anchor	Meaning here
L (left)	Orders with **NO** matching return = **NOT returned** orders
J (join)	Orders that **DID** match a return = **Returned** orders (this output has the `Status` column)
R (right)	Returns with no matching order = **ignore these**

TASK 4 — Union "not returned" (L) with "returned" (J); ignore R (≈10 marks)

1. Tool Palette → **Join** category → drag the **Union** tool onto the canvas.
2. Connect **Join's L output → Union input.**
3. Connect **Join's J output → Union input.** (Two arrows now feed the Union. **Do NOT connect R.**)
4. Click Union → Configuration → set **"Auto Config by Name"** (matches columns by their names).
5. **Ctrl+R** to Run.

Check: You now have ALL orders again (returned + not returned). The `Status` column shows **"Returned"** on the matched rows and **[Null]** on the not-returned rows. Those nulls are what Task 5 fixes.

> **Why this works:** The L rows never had a `Status` field, so Union fills it with null. The J rows carry `Status = "Returned"`. Stacking them = every order, correctly labelled or blank.

TASK 5 — Fill null STATUS with "Not Return" (≈10 marks)

1. Tool Palette → **Preparation** category → drag the **Formula** tool onto the Union's output.
2. Click Formula → Configuration → **+ Select Column →** `Status` (you are updating the existing Status field, not creating a new one).
3. In the expression box type **exactly**: `IF IsNull([Status]) THEN "Not Return" ELSE [Status] ENDIF`
4. **Ctrl+R** to Run.

Check: No more blanks in `Status`. Every row is either **"Returned"** or **"Not Return"**.

> Match the paper's wording exactly: **`Not Return`** (not "Not Returned"). Little things score marks.

TASK 6 — Save output as `ProductOrders20102013.hyper` (≈2 marks)

`.hyper` is **Tableau's** data format — you export it here so Section C can load it.

1. Tool Palette → **In/Out** → drag **Output Data** onto the Formula's output.
2. Configuration → **Write to File or Database** → browse to **your exam folder**.

3. In the file-name box, set the **"Save as type"** dropdown to **Tableau Hyper (*.hyper)**.
4. Name it **exactly**: `ProductOrders20102013.hyper`
5. **Ctrl+R** to Run one final time.

Done. Confirm the `.hyper` file now sits in your exam folder. **Ctrl+S** the workflow. Section B complete.

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4. Final picture your canvas should match

```
 Input(ProductOrders_*.xlsx)──┐
                              ├─[Join]  L ─┐
Input(Product_Returned.xlsx)──┘        J ──┴─[Union]─[Formula: Status]─[Output .hyper]
                                       R  (not connected)
```

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5. Rehearse-it-blind checklist (tick without looking at steps)

- [ ] Made matric-name folder, saved workflow into it
- [ ] One Input Data with `ProductOrders_*.xlsx|||Sheet1`
- [ ] Second Input Data = `Product_Returned.xlsx`
- [ ] Join: Left=Orders, Right=Returns, key = **Order ID**
- [ ] Union: connect **L + J only** (never R), Auto Config by Name
- [ ] Formula on **Status**: `IF IsNull([Status]) THEN "Not Return" ELSE [Status] ENDIF`
- [ ] Output Data → **Tableau Hyper** → `ProductOrders20102013.hyper`
- [ ] Ran the whole thing (Ctrl+R), file exists, Ctrl+S

Do this workflow 3 times today from a blank canvas. By the third time it takes ~6 minutes and your hands know it. That is how you walk into Section B calm.

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6. Panic buttons (if something breaks)

Problem	Fix
Wildcard won't read all files	Use 4 Inputs → Union (fallback above). Partial marks > zero.
Join gives 0 matched rows	Check both keys are **Order ID** and same type. Add a **Select** tool to make both Order ID = Int64 if types differ.
`Status` field missing after Union	You unioned the wrong anchors, or used "by Position". Set Union to **Auto Config by Name**; make sure **J** is connected.
Can't find `.hyper` in Save type	Type the full path ending `...\ProductOrders20102013.hyper` and pick Hyper in the format dropdown; if absent, output `.yxdb` or `.csv`, still submit, and tell no one it wasn't your first choice.
Everything red	Read the error at the bottom — it names the tool. Click that tool, fix the field it complains about, re-Run.

> **Marks mindset:** every task is *independently* graded. Even if Task 3 stumbles, do Tasks 1, 2, 5, 6 anyway. Never leave a task blank.

SECTION C — TABLEAU (≈40 marks)

Zero-to-Hero Walkthrough — SECP3213 Business Intelligence Final

> **What this is:** All 10 past-paper Tableau questions solved click-by-click, using the `.hyper` file you exported in Section B. Every technique here is one your lecturer taught in the CH05 lab — I've matched their exact menu names. One worksheet per question, renamed **Q1, Q2, …** (the paper requires this).

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0. The 60-second mental model

Tableau = drag **fields** onto **shelves** to make a chart. Two field types: - **Dimensions** (blue) = categories/text/dates → *State, Region, Product Name, Order Date, Status.* - **Measures** (green) = numbers Tableau adds up → *Sales, Profit, Shipping Cost, Quantity.*

You build a chart by putting dimensions on **Rows/Columns/Filters** and measures into the view. To get an **answer**, you read the number/label off the chart or hover a mark.

5 zones to know: | Zone | Where | Use | |---|---|---| | **Data pane** | Left (fields list) | Grab fields | | **Columns / Rows** shelves | Top bars | Structure the chart | | **Marks card** (Color, Size, Label, Detail, Text) | Left-middle | Add fields to marks | | **Filters card** | Above Marks | Limit the data | | **Analytics pane** | Tab next to Data (top-left) | Forecast, CI, trend, average lines |

Every question = new worksheet. Bottom-left → click the "new worksheet" tab icon. Double-click its tab to **rename to Q1/Q2/…**. **Ctrl+S** constantly.

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1. Load the `.hyper` from Section B (do first)

1. Open **Tableau Desktop**.
2. Connect pane (left) → **To a File** → **More…** → browse to your exam folder → select `ProductOrders20102013.hyper` → Open. *(If you don't see it: To a File → the type may show as "Tableau Hyper" / "Extract" — pick the .hyper.)*
3. On the Data Source page, if the table isn't already shown, drag the table (**Extract** / `ProductOrders...`) into the canvas ("Drag tables here").
4. Bottom-left → click **Sheet 1**. You're ready.
5. **File → Save As** → save as `SectionC.twbx` (Packaged Workbook) into your matric folder.

> Sanity check your fields exist: **Sales, Profit, Shipping Cost, Order Date, Region, State or Province, Product Category, Product Sub-Category, Product Name, Product Container, Customer ID, Order ID, Status.** If a number came in as text, click its data-type icon in the Data Source page and fix it (e.g., Sales → Number/decimal, Order Date → Date).

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2. THE 10 QUESTIONS

Q1 — % of sales for items shipped in a "Medium Box", for Jan 2010–2013 (6 marks)

> "Medium Box" = the **Product Container** field (not Ship Mode).

1. New worksheet → rename **Q1**.
2. **Product Container** → **Rows**.
3. **Order Date** → **Columns**. Click the pill's ▾ → choose **Year** (discrete). You get columns 2010–2013.
4. **Order Date** → **Filters** → choose **Months** → **Next** → tick **January** only → OK. *(Now only January of each year is counted.)*
5. **Sales** → drop onto **Text/Label** in the Marks card (or into the table).
6. Right-click the **SUM(Sales)** pill → **Quick Table Calculation** → **Percent of Total**.
7. Right-click it again → **Compute Using** → **Table (Down)** so each year's column totals 100%.
8. Marks → **Label** → tick **Show mark labels**.

Answer = the % shown on the "Medium Box" row for each year (2010, 2011, 2012, 2013).

Q2 — State with lost (negative) profit in BOTH 2012 and 2013 (4 marks)

1. New worksheet → **Q2**.
2. **State or Province** → **Rows**.
3. **Order Date** → **Columns** → set pill to **Year**.
4. **Order Date** → **Filters** → **Years** → tick **2012** and **2013** → OK.
5. **Profit** → **Color** on Marks (red/orange = negative). Also drag **Profit** → **Label** to read values.
6. Scan for the state that is **negative in both** the 2012 and 2013 columns. *(Tip: right-click State → Sort → by Profit ascending to float losses to the top.)*

Answer = the state negative in both years.

Q3 — Region + Product Category + Sub-Category that lost profit (negative) (5 marks)

1. New worksheet → **Q3**.
2. **Region** → **Rows**, then **Product Category** → **Rows** (right of Region), then **Product Sub-Category** → **Rows** (right again). Nested breakdown.
3. **Profit** → **Columns** (bars appear).
4. Right-click the **Sub-Category** pill (innermost) → **Sort** → **Field / Profit / Ascending**.
5. **Profit** → also drop on **Label**. The bars going **left/negative** are your losers.

Answer = every Region › Category › Sub-Category combination with negative profit (read the negative bars).

Q4 — Average + upper & lower bound at 95% CI of Sales, 2010–2013 (5 marks)

1. New worksheet → **Q4**.
2. **Order Date** → **Columns** → set to **Year**.
3. **Sales** → **Rows** (four bars/marks, one per year).
4. Top-left → click the **Analytics** tab (next to Data).
5. Under **Model**, drag **"Average with 95% CI"** onto the view → drop on **Table**.
6. Hover the average line and the shaded band edges to read the three numbers.

Answer = Average = (line value); Upper bound & Lower bound = the top/bottom of the shaded 95% band.

> If "Average with 95% CI" isn't listed: drag **Average Line** (gives the average), then Analytics → **Distribution Band** → Value → **Confidence Interval 95%** for the bounds.

Q5 — Forecast Sales for 2014 (5 marks)

1. New worksheet → **Q5**.
2. **Order Date** → **Columns**. Click the pill ▾ → pick the **continuous** (green) **Year** — choose the second "Year" entry so the axis is continuous, giving a line.
3. **Sales** → **Rows** (a line across 2010–2013).
4. **Analytics** tab → under **Model**, drag **Forecast** onto the view. Tableau extends a shaded forecast.
5. Right-click the forecast → **Forecast** → **Forecast Options** → set **Forecast Length** → **Exactly 1 Years** → OK.
6. Hover the **2014** point to read the forecasted value.

Answer = the forecasted Sales value at 2014.

Q6 — Scatter: product with high Shipping Cost but low Sales (3 marks)

1. New worksheet → **Q6**.
2. In the Data pane, **Ctrl-click** to highlight **Shipping Cost** and **Sales** → click **Show Me** (top-right) → pick **scatter plot**.
3. Menu **Analysis** → untick **Aggregate Measures** (spreads points out).
4. **Product Name** → **Detail** on Marks (each dot = a product). **Product Category** → **Color**.
5. Make sure axes are: **Sales** (one axis) vs **Shipping Cost** (other). Look for the dot that is **high on Shipping Cost but far left/low on Sales** → hover it.

Answer = that product (Category › Product Name).

Q7 — Products (by Product Name) returned more than once (2 marks)

1. New worksheet → **Q7**.
2. **Status** → **Filters** → tick **Returned** → OK.
3. **Product Name** → **Rows**.
4. **Order ID** → drag into view → right-click the pill → **Measure** → **Count (Distinct)**. *(= number of returned orders per product.)*

5. Right-click Product Name → **Sort** → by that field, **Descending**. Optionally drag the count to **Label**.
6. Read products whose count is **2 or more**.

Answer = the product names with returned-count > 1.

Q8 — How many orders in East region were returned (2 marks)

1. New worksheet → **Q8**.
2. **Region** → **Filters** → tick **East** → OK.
3. **Status** → **Filters** → tick **Returned** → OK.
4. **Order ID** → into view → right-click pill → **Measure → Count (Distinct)**.
5. Marks → **Label** on (or just read the single number).

Answer = that count.

Q9 — Region with highest SUM(Shipping Cost) for returned orders (3 marks)

1. New worksheet → **Q9**.
2. **Status** → **Filters** → **Returned** → OK.
3. **Region** → **Rows**.
4. **Shipping Cost** → **Columns** (SUM by default). Drag it to **Label** too.
5. Sort **descending** (toolbar sort-icon). Top row = answer.

Answer = the top region + its shipping-cost value.

Q10 — Customers in 2012 who also purchased in 2010 (5 marks)

> This is the "Combined Set" technique from the lab (Analysis 12).

1. New worksheet → **Q10**.
2. **Customer ID** → **Rows**. (Customer ID is safer than Name; either works.)
3. **Order Date** → **Filters** → **Years** → tick **2010** → OK.
4. Click in the view, press **Ctrl+A** to select all customers → right-click the selection → **Create Set** → name it **Customers 2010** → OK.
5. Filters card → right-click **Order Date** → **Edit Filter** → change to **2012** → OK.
6. Press **Ctrl+A** again → right-click → **Create Set** → name **Customers 2012** → OK.
7. Data pane → hold **Ctrl**, click **both** sets (Customers 2010 + Customers 2012) → right-click → **Create Combined Set** → name **2010 & 2012** → choose **"Shared members in both sets"** → OK.
8. Clear the sheet (toolbar **Clear ▾ → Clear Sheet**).
9. **Customer ID** → **Rows** → right-click the pill → **Measure → Count (Distinct)**.
10. Drag the **2010 & 2012** combined set → **Filters** (keep **In**). Marks → **Label** on.

Answer = the single count shown = customers who bought in both 2010 and 2012.

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3. Save & submit

1. Confirm 10 tabs: **Q1 … Q10**.
2. **File → Export Packaged Workbook…** (or Save As → type **Tableau Packaged Workbook .twbx**) → save into your **matric-name folder**.
3. Leave nothing blank — a half-built worksheet still scores partial marks.

4. Technique cheat-sheet (memorize these triggers)

You need…	Do this
A **percentage**	Right-click the measure → Quick Table Calculation → **Percent of Total** (+ Compute Using)
Negative profit spotting	Profit → **Color**; or Sort ascending
Average + 95% CI	Analytics → Model → **Average with 95% CI**
Forecast a future year	Analytics → Model → **Forecast** → Forecast Options
Scatter	highlight 2 measures → Show Me → scatter → Analysis → untick **Aggregate Measures**
Count of things	field → right-click → Measure → **Count (Distinct)**
"Same customers in two years"	Make two **Sets**, then **Create Combined Set** → *Shared members*
A **map**	double-click **State** → drag Profit to Color
Filter to a value	drag field → **Filters** → tick value

5. Panic buttons

Problem	Fix
`.hyper` won't load	Connect → To a File → **More…** → change file type filter to *All*/Hyper → pick it.
A number acts like text (can't SUM)	Data Source page → click the field's data-type icon → **Number (decimal)**.
Filter shows wrong values	You filtered the wrong field. Remove the pill from Filters, re-drag the correct one.
"Percent of Total" all show 100%	Right-click measure → **Compute Using** → pick the right dimension (Table Down/Across).
Forecast/Analytics greyed out	The date must be **continuous** (green). Re-drop Order Date and pick the green Year.
Map is blank / "unknown"	Click the "X unknown" bottom-right → Edit Locations → set Country.

> **Golden rule:** one worksheet per question, **renamed Q#**, answer visible as a label. Do 3–4 of these tonight until the moves feel automatic; the exam reuses the same handful of techniques.

SECTION A — THEORY CRIB SHEET (≈10 marks, 20 min, NO laptop)

SECP3213 Business Intelligence — Topics 1–5

> **How to use this:** Section A is short — 20 minutes, answer booklet, likely MCQ + a few short structured questions. You do **not** need deep essays; you need crisp definitions and the ability to pick/justify the right term. Memorize the **bold** terms and the **past-paper answer key** at the end. That answer key alone historically covers most of Section A.

TOPIC 1 — Introduction to Business Intelligence

BI definition (one-liner to memorize): BI is a set of *technologies, applications and processes* that **collect, integrate, analyze and present** business data to **support better decision-making**. Essentially a **data-driven Decision Support System (DSS)**.

Why BI matters: improves access to data → turns data into **insights** → better strategic/tactical/operational **decisions** → higher productivity, revenue, competitive advantage.

Evolution: | Era | Milestone | |---|---| | 1980s | EIS, DSS | | 1990s | Data Warehousing, BI | | 2000s | Dashboards, Scorecards, Performance Mgmt | | 2010s+ | Analytics, Big Data, Self-service/Personal BI, AI/ML |

Big Data — the 5 V's: Volume, Velocity, Variety, Veracity, Value.

Data types: structured (~15%) vs unstructured (~85%); internal vs external; numeric vs textual.

Decision-making steps: 1) Identify problem → 2) Seek info on options & effects → 3) Evaluate & choose → 4) Implement → 5) Monitor & adjust.

Report vs Dashboard (common exam contrast): | Report | Dashboard | |---|---| | Detailed data, tabular, numbers/text | Visual, most-important data at a glance | | Based on simple queries; for printing/export | Consolidated on one screen to monitor goals |

BI Governance: processes/policies ensuring data is **trusted, secure, high-quality, compliant**, and consistently used across the org. (Data governance = the data itself; **BI governance** extends this to all report/dashboard consumers.)

TOPIC 2 — Components of BI Architecture

BI Architecture = blueprint of technology + processes + policies that **collect → store → analyze** data for decisions.

Core components (memorize the pipeline): 1. **Data Sources** — operational/source systems (ERP, CRM, transaction systems), plus external (social, sensors/RFID, market data). 2. **Data Integration (ETL/ELT)** — move, clean, organize data. 3. **Data Storage** — data warehouse / data lake / data marts. 4. **Semantic Layer** — maps complex data into **business-friendly terms**. 5. **Analytics & Reporting Layer** — dashboards, visualization. 6. **Governance & Security** — trust, compliance.

ETL vs ELT: Extract, **T**ransform, **L**oad (transform before loading). **ELT** = load first, transform in the warehouse. **Transform** is the most time-consuming step (fixing inconsistent codes, incomplete data).

Operational/Embedded BI: when BI is built into an operational task (e.g., "out of stock → suggest alternatives"). The nugget shown = **embedded BI**.

Metadata = "data about data". Two kinds: - **Technical metadata** — where data came from, when extracted/loaded. - **Business metadata** — business meaning & calculations (e.g., revenue = price × qty − discounts).

Master Data Management (MDM): consistent definitions of core entities (customer, product) across all systems (one product ID everywhere).

Data Warehouse vs Data Mart: | Data Warehouse | Data Mart | |---|---| | Enterprise-wide, integrated, subject-oriented | Subset for one **business unit/function** | | Serves whole org | Serves a specific need |

Two DW design paradigms: - **Inmon (top-down):** one enterprise DW in 3rd normal form; marts source from it. - **Kimball (bottom-up):** DW = collection of data marts; **dimensional model**.

Medallion architecture: Bronze (raw) → **Silver** (cleaned) → **Gold** (business-ready).

Warehouse tables: - **Fact table** = numeric **measures** (sales, profit, quantity) + keys. Fine **granularity** = very detailed rows. - **Dimension table** = descriptive context (who/what/where/when) — product, time, geography. Also called lookup/reference tables.

Schemas: | Schema | Shape | |---|---| | **Star** | 1 fact table + surrounding dimension tables (denormalized) | | **Snowflake** | Star with dimensions **normalized** into sub-tables | | **Fact constellation / galaxy** | Multiple fact tables sharing dimensions |

Cloud BI models: IaaS (rent infrastructure) → **PaaS** (infrastructure + platform software) → **SaaS** (ready-to-use software, e.g., subscribe & upload data). **DW appliance** = server + database + storage in one optimized box.

TOPIC 3 — BI Front-End

Self-Service BI: lets business users access/analyze data **without IT help** — filter, sort, visualize on their own. Reduces IT backlog.

Business Query & Reporting: users build queries/reports **without writing SQL**, via a **business view / metadata layer** that hides physical DB complexity (business terms, auto-joins, pre-built metrics). - **Ad-hoc report** = one-off, exploratory question. - **Fixed report** = recurring, standardized.

Visual Data Discovery: speeds **time-to-insight** through visualization & easy exploration; moves focus from **"what"** to **"why/where/when"**. Tools: **Tableau**, Spotfire, Qlik, SAP Lumira. Often uses in-memory processing + mashing multiple sources.

Dashboard: visual display of the most important info on **one screen**, monitored **at a glance**; pulls from **multiple data sources** (map + trend line + KPI + cross-tab).

Scorecard vs Dashboard: | Scorecard | Dashboard | |---|---| | Tracks **KPIs vs target/forecast** | Presents multiple numbers many ways | | Strategic; **Balanced Scorecard** = People, Customers, Financial, Operations | Operational/monitoring, traffic-light metrics |

KPI (Key Performance Indicator): a **quantifiable metric** evaluating performance **against an objective/target**.

OLAP (Online Analytical Processing): multidimensional analysis of data (the "cube"). Operations: **drill-down / roll-up, slice / dice, pivot**. Contrast with **OLTP** (Online Transaction Processing = day-to-day transactions).

Descriptive statistics (business analytics basics): - **Central tendency:** Mean, Median, Mode. - **Dispersion:** Range, Variance, Standard Deviation, MAD; **Quartiles / Box-plot**. - **Shape:** Skewness (asymmetry), Kurtosis (peakedness). - **Descriptive** (describe the sample) vs **Inferential** (infer about the population).

TOPIC 4 — Data Analytic Applications (Alteryx)

The 4 types of analytics (VERY common MCQ): | Type | Question it answers | Example | |---|---|---| | **Descriptive** | *What happened?* | Sales report last quarter | | **Diagnostic** | *Why did it happen?* | Drill into a sales drop | | **Predictive** | *What will happen?* | Forecast next-year sales | | **Prescriptive** | *What should we do?* (best action) | Recommend optimal pricing |

Data preprocessing (why): "garbage in, garbage out." Steps: **cleaning** (missing values, duplicates, outliers), **integration**, **transformation** (normalize, format), **reduction**.

Alteryx ETL concepts (map to Section B): - **Input Data** (read files; wildcard `*` for many files) → **Select** (rename/retype) → **Filter** (condition → True/False) → **Formula** (create/update fields) → **Join** (combine on key: L/J/R outputs) → **Union** (stack datasets) → **Summarize** (group & aggregate) → **Output Data** (write result). - **Data Investigation** tools = field summary, association analysis, distribution (EDA before modeling).

Predictive analytics techniques: - **Regression** — predict a **numeric** value (e.g., sales). Linear regression fits a line. - **Classification** — predict a **category/label** (e.g., churn: yes/no). Decision trees, logistic regression. - **Clustering** — **unsupervised** grouping of similar records (e.g., customer segments, k-means). No labels.

Lift chart / confidence interval — evaluate model quality; CI expresses the range the true value likely falls within (e.g., **95% CI**).

TOPIC 5 — Building Visual Models (Tableau & Power BI)

Why visual models: humans process visuals faster → quicker insight, pattern/outlier spotting, storytelling.

Choosing the right chart: | Goal | Chart | |---|---| | Trend over time | **Line** | | Compare categories | **Bar** | | Relationship between 2 measures | **Scatter** | | Part-to-whole | **Pie / Treemap / Stacked bar** | | Distribution | **Histogram / Box plot** | | Geographic | **Map** | | Intensity across 2 dims | **Heat map** |

Visualization best practices: right chart for the data; minimize clutter (high data-ink ratio); consistent, meaningful color; clear labels/titles; avoid misleading axes; design for the audience (**UX**).

Tableau essentials: - **Dimensions** (blue, qualitative) vs **Measures** (green, quantitative, auto-aggregated). - **Discrete** (blue, separate values) vs **Continuous** (green, any value in range). - Connection: **Live** (real-time, reflects source changes) vs **Extract** (`.hyper` snapshot, works offline, faster). - Save formats: **.twb** (workbook, no data) vs **.twbx** (packaged, includes data).

Power BI essentials (concept only — not in the hands-on exam): - Workspace: **Power Query** (import & transform, ETL) → **Data model** → **Report** visuals. - **DAX** = formula language for measures/calculated columns. - **Slicers/Filters** for interactivity; publish to Power BI Service.

Tableau vs Power BI (one-liner): Tableau = best-in-class visualization/exploration; Power BI = tightly integrated with Microsoft, cheaper, DAX-driven.

★ PAST-PAPER SECTION A — ANSWER KEY (memorize these)

From the 2023/24 final. Expect the same or very similar questions.

#	Question topic	Answer	Why
1	Primary purpose of a **data warehouse**	**(c)** Centralized repository for structured, organized data	DW = integrated central store for analysis
2	Analytics that suggests **best course of action**	**(c)** Prescriptive analytics	Prescriptive = "what should we do"
3	Role of a **KPI**	**(c)** Quantifiable metric for evaluating performance vs objectives	Definition of KPI
4	**OLAP** stands for	**(b)** Online Analytical Processing	—
5	Significance of **data governance**	**(b)** Safeguarding data quality, security, compliance	—
6	Technique analyzing **textual data** for patterns	**(a)** Sentiment analysis	Text analytics on unstructured text
7	Primary function of a **BI query tool**	**(c)** Let users request specific info from the database	Query = ask the data (no SQL)
8	How BI gives **competitive advantage**	**(c)** Provides actionable insights competitors lack	—
9	Purpose of a **DataMart**	**(b)** Subset of data specific to a business function/team	Mart = departmental subset
10	Role of **data granularity**	**(c)** Defines the level of detail captured in the data	Fine grain = more detail

Pattern: the answer is almost always the *complete, insight/decision-oriented* option; the wrong answers are usually "storing only raw data", "restricting access", "deleting data", or overly narrow.

STRUCTURED-ANSWER BANK (if they ask "explain/define" — write ~2–3 sentences)

- **Define BI:** *Technologies and processes that collect, integrate, analyze and present data to support better business decisions; a data-driven DSS.*
- **ETL:** *Extract data from sources, Transform (clean/standardize) it, Load into the data warehouse. Transform is the most time-consuming step.*

- **Star vs Snowflake:** *Star = one central fact table with denormalized dimension tables (simple, fast). Snowflake = dimensions normalized into sub-tables (less redundancy, more joins).*
- **Fact vs Dimension table:** *Fact holds numeric measures + keys; Dimension holds descriptive context (who/what/where/when) used to filter and label.*
- **Dashboard vs Scorecard:** *Dashboard monitors many metrics at a glance from multiple sources; Scorecard tracks KPIs against strategic targets.*
- **Descriptive→Diagnostic→Predictive→Prescriptive:** *What happened → why → what will happen → what to do.*
- **Regression vs Classification vs Clustering:** *Regression predicts a number; Classification predicts a category (both supervised); Clustering groups similar records without labels (unsupervised).*
- **Self-service BI:** *Lets business users access and analyze data themselves without IT, reducing backlog.*
- **Metadata:** *Data about data — technical (source/lineage) and business (meaning/calculations).*
- **Live vs Extract (Tableau):** *Live = real-time link to source; Extract (.hyper) = a fast, offline snapshot.*

> **20-minute strategy:** answer every MCQ (no blanks — no negative marking). For structured items, lead with the definition, then one example. Keep it tight; you have a laptop war to fight right after.

(Mock paper for practice — not an official UTM document.)

CONFIDENTIAL

UNIVERSITI TEKNOLOGI MALAYSIA — FACULTY OF COMPUTING

FINAL EXAMINATION (MOCK) — SEMESTER 2, 2025/2026

SUBJECT CODE: SECP3213
SUBJECT NAME: BUSINESS INTELLIGENCE
TIME: 3 HOURS
INSTRUCTIONS: Section A on the answer booklet (no laptop). Sections B & C on your laptop; save files in a folder named `MATRIC_FULLNAME`.

NAME	MATRIC NO.	SECTION	LECTURER

SECTION A [10 MARKS] — Circle the correct answer.

1. Business Intelligence is essentially a data-driven form of which system? a) Online Transaction Processing (OLTP) b) Decision Support System (DSS) c) Enterprise Resource Planning (ERP) d) Relational Database Management System (RDBMS)

2. Which of the following is **NOT** one of the 5 V's of Big Data? a) Volume b) Variety c) Value d) Validity

3. In an **ELT** pipeline, the *Transform* step is performed: a) before extraction b) before loading c) after loading, inside the data warehouse d) only during extraction

4. A schema in which dimension tables are **normalized** into smaller sub-tables is called a: a) Star schema b) Snowflake schema c) Fact table d) Operational schema

5. Which table type stores numeric measures such as *Sales* and *Profit*? a) Dimension table b) Lookup table c) Reference table d) Fact table

6. Analytics that answers the question **"Why did it happen?"** is: a) Diagnostic analytics b) Descriptive analytics c) Predictive analytics d) Prescriptive analytics

7. The primary benefit of **self-service BI** is that it: a) eliminates the need for a data warehouse b) stores only raw, unprocessed data c) lets business users access and analyze data without IT involvement d) replaces all dashboards with printed reports

8. A **scorecard** differs from a dashboard because it: a) can only display maps b) tracks KPIs against a target or forecast c) is unable to display any KPI d) is always an operational, real-time tool

9. Which predictive technique is **unsupervised** (uses no labelled outcome)? a) Linear regression b) Classification c) Logistic regression d) Clustering

10. A Tableau `.hyper` **extract** is best described as: a) a fast, offline snapshot of the data b) a live connection to the source system c) a workbook that contains no data d) a stored SQL query

---

SECTION B [50 MARKS] — Alteryx Designer

Design a workflow in **Alteryx Designer** using the provided datasets.

1. Read **all four** order datasets using **one** Input Data tool: `ProductOrders_2010.xlsx`, `ProductOrders_2011.xlsx`, `ProductOrders_2012.xlsx`, `ProductOrders_2013.xlsx`. **(8 marks)**

2. Add a second Input Data tool to read `Product_Returned.xlsx`. **(5 marks)**

3. Join the product-orders dataset (Left) with the product-returned dataset (Right) on **Order ID**. **(12 marks)**

4. Union the product orders that were **not returned** (L anchor) with the product orders that **were returned** (J anchor). **Ignore** the data from the R anchor. **(8 marks)**

5. Using a **Formula** tool, update the field **STATUS**: where it is null, set it to **"Not Return"**. **(7 marks)**

6. Add a **Summarize** tool to compute the **SUM of Sales**, grouped by **Region** and **STATUS**. **(5 marks)**

7. Save the joined & cleaned dataset (the output *before* the Summarize) as `ProductOrdersClean.hyper`. **(5 marks)**

SECTION C [40 MARKS] — Tableau Desktop

Create a workbook in **Tableau Desktop**. Load `ProductOrdersClean.hyper` from Section B. **Each answer on its own worksheet, renamed by question number (e.g., Q1).**

1. Find the **percentage of total Sales** contributed by each **Ship Mode** in year **2011**. **(5 marks)**

2. Which **Region** has the **highest total Profit** across 2010–2013? **(3 marks)**

3. List **every Product Sub-Category** that has **negative total Profit**. **(5 marks)**

4. Determine the **average Sales** with the **upper and lower bound at 95% Confidence Interval** for 2010–2013. **(5 marks)**

5. Forecast the total **Sales for 2015** (two years beyond the data). **(5 marks)**

6. Draw a **scatter plot** to identify which **product (Product Name)** has **high Quantity ordered but low Profit**. **(4 marks)**

7. How many **distinct customers** made a purchase in **2013**? **(3 marks)**

8. How many **orders in the West region** were **returned**? **(3 marks)**

9. Which **Product Container** has the **highest total Shipping Cost** for **returned** orders? **(4 marks)**

10. How many customers who purchased in **2010** also purchased in **2011**? **(3 marks)**

End of Mock Questions

> **Self-test protocol:** Section A closed-book in 15 min. Then Sections B & C on your laptop in ≤2h30. Only after finishing, open `05_Answer_Scheme.md` to grade yourself.

ANSWER SCHEME — Predicted Mock Exam (SECP3213)

! Do the mock first. Grade yourself honestly against this.

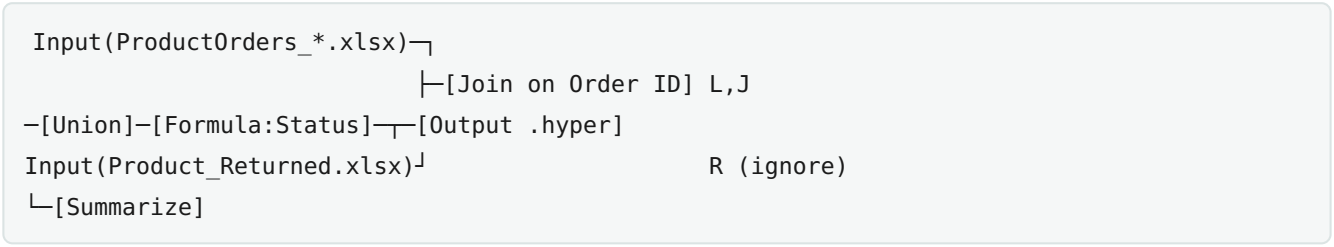
SECTION A — MCQ KEY [10 marks, 1 each]

#	Ans	Why
1	**B**	BI = a data-driven **Decision Support System**.
2	**D**	The 5 V's are Volume, Velocity, Variety, Veracity, Value. "Validity" is not one.
3	**C**	ELT = Extract, **Load**, then **Transform inside the warehouse**.
4	**B**	Snowflake = star schema with **normalized** dimensions.
5	**D**	**Fact table** holds numeric measures (Sales, Profit).
6	**A**	"Why did it happen" = **Diagnostic**.
7	**C**	Self-service BI = users analyze data **without IT**.
8	**B**	Scorecard = **KPIs vs targets**.
9	**D**	**Clustering** is unsupervised (no labels).
10	**A**	`.hyper` extract = **offline snapshot** of the data.

Letter distribution: A×2, B×3, C×2, D×3 (balanced). **Your target: 10/10 — these are all in the crib sheet.**

SECTION B — Alteryx model answer [50 marks]

Canvas:



Q1 — One Input, all 4 files (8): Input Data → pick `ProductOrders_2010.xlsx`, sheet `Sheet1` → edit path to `ProductOrders_*.xlsx|||Sheet1`. [+4 one Input tool used] [+4 wildcard reads all four years]

> **Trap:** hard-coding only 2010, or using 4 separate inputs → loses the "one tool" marks. Wildcard is the point.

Q2 — Second Input (5): Input Data → `Product_Returned.xlsx` → Sheet1. [+5]

> **Trap:** appending it to the first input instead of a separate tool.

Q3 — Join on Order ID (12): Join tool; **Left = orders**, **Right = returns**; key **Order ID = Order ID**. [+4 correct tool] [+4 Left/Right correct] [+4 join key = Order ID]

> **Trap:** swapping Left/Right — then the L anchor stops meaning "not returned" and Q4 breaks. Orders MUST be Left.

Q4 — Union L + J, ignore R (8): Union tool fed by **L and J only**; Auto Config by Name. [+3 L connected] [+3 J connected] [+2 R correctly ignored]

> **Trap:** connecting R, or using "by Position" so columns misalign.

Q5 — Formula fixes null STATUS (7): Formula → update `STATUS`:

```
IF IsNull([STATUS]) THEN "Not Return" ELSE [STATUS] ENDIF
```

[+3 correct tool + updates existing STATUS] [+2 IsNull logic] [+2 exact string "Not Return"]

> **Trap:** creating a NEW field instead of updating STATUS; or typing "Not Returned" (extra *d*).

Q6 — Summarize SUM(Sales) by Region & STATUS (5): Summarize → Group By **Region**, Group By **STATUS**, **Sum → Sales**. [+2 two group-bys] [+3 Sum of Sales]

> **Trap:** grouping by Region only, or using Count instead of Sum.

Q7 — Output `.hyper` (5): Output Data on the Formula output (before Summarize) → type **Tableau Hyper (*.hyper)** → `ProductOrdersClean.hyper`. [+3 correct .hyper format] [+2 exact filename]

> **Trap:** exporting the *summarized* stream (you'd lose row-level data Section C needs). Output the cleaned rows.

SECTION C — Tableau model answers [40 marks]

> Exact numbers depend on your build — **the method is what earns marks.** Each on its own worksheet, renamed Q1…Q10, answer visible as a label.

Q1 — % of Sales by Ship Mode, 2011 (5): Ship Mode→Rows; Sales→Text; filter Order Date **Year = 2011**; right-click SUM(Sales) → **Quick Table Calc → Percent of Total**. [+2 filter 2011] [+2 % of total] [+1 by Ship Mode] · **Trap:** forgetting the year filter → % over all years.

Q2 — Region with highest Profit (3): Region→Rows; SUM(Profit)→Columns; sort **descending**; top = answer. [+2 correct build] [+1 reads top] · **Trap:** filtering out years — question wants all 2010–2013.

Q3 — Sub-Categories with negative Profit (5): Product Sub-Category→Rows; SUM(Profit)→Columns; sort ascending; **all negative bars** are the answer. [+3 build] [+2 lists all negatives] · **Trap:** naming only one — list **every** negative sub-category.

Q4 — Average Sales + 95% CI (5): Year→Columns, Sales→Rows; Analytics → Model → **Average with 95% CI**; hover for average + upper/lower bounds. [+2 average line] [+3 upper & lower CI bounds] · **Trap:** giving only the average, not the two bounds.

Q5 — Forecast Sales 2015 (5): Order Date (continuous Year)→Columns; Sales→Rows; Analytics → **Forecast**; Forecast Options → length **2 years**; read 2015. [+3 forecast added] [+2 extends to 2015] · **Trap:** date is discrete (blue) → forecast greyed out. Must be continuous (green).

Q6 — Scatter: high Quantity, low Profit (4): highlight **Quantity + Profit** → Show Me → scatter; Analysis → untick **Aggregate Measures**; **Product Name**→Detail; find high-quantity/low-profit dot. [+2 scatter] [+1 disaggregate] [+1 identifies product] · **Trap:** leaving measures aggregated → one dot.

Q7 — Distinct customers in 2013 (3): filter Year=2013; **Customer ID → Count (Distinct)** → Label. [+2 COUNTD] [+1 filter 2013] · **Trap:** using Count (not Distinct) → counts orders, not customers.

Q8 — Returned orders in West (3): filters **Region=West** and **STATUS=Returned**; **Order ID → Count (Distinct)**. [+2 both filters] [+1 count] · **Trap:** forgetting one filter.

Q9 — Container with highest Shipping Cost, returned (4): filter STATUS=Returned; Product Container→Rows; SUM(Shipping Cost)→Columns; sort desc. [+2 filter returned] [+2 correct container + value] · **Trap:** using Ship Mode instead of Product Container.

Q10 — Customers in both 2010 & 2011 (3): make Set "Customers 2010" and Set "Customers 2011" (filter year → Ctrl+A → Create Set); **Create Combined Set → Shared members in both**; Customer ID → COUNTD, filter by combined set. [+2 two sets + combined "shared"] [+1 count] · **Trap:** choosing "All members of both sets" (union) instead of **shared members** (intersection).

COVERAGE MAP (this mock → your topics)

Topic	Where tested
1 Intro to BI	A1, A2
2 BI Architecture	A3, A4, A5
3 Front-End (self-service, scorecard)	A7, A8
4 Analytics types + Alteryx ETL	A6, A9, **all of Section B**
5 Visual models + Tableau	A10, **all of Section C**

Every topic is hit. Section B and C carry 90% of the marks → **that's where your hours go.**

TRAINING PROTOCOL (how to use this mock)

1. **Round 1 (today):** Do Section A closed-book. Do B & C *with* the walkthroughs open. Goal: understand every click.
2. **Round 2 (tonight):** Redo B & C from a blank canvas, walkthroughs **closed**. Time yourself.
3. **Round 3 (exam morning):** Do B & C blind in < 45 min total. If you can, you're ready.
4. Re-take Section A MCQs until 10/10 without peeking.

Scoring yourself: B and C are graded per-task and per-worksheet — partial credit everywhere. Never leave anything blank; a half-right worksheet still scores.